



ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

Course Number and Title: ROB 2040 Programmable Controller Applications

Section Number: A1501

Course Start and End Dates: 5/5/2025 – 8/18/2025

Meeting Days and Times: Monday, 5:00pm – 9:50pm

Classroom Location: T4

Term and Year: Summer 2025

Instructor Name: Jason Workman

E-Mail: jrworkma@oaklandcc.edu

Textbook: None

Other needed supplies: Flash Drive

Course Description: This class will introduce the fundamentals of Programmable Logic Controller (PLC) operations, including symbology and programming techniques. PLC hardware and data structures will be presented. Methods of using the programming interface to troubleshoot applications will be emphasized. The student will write, enter, and execute application programs using the programmable controllers and Human Machine Interface (HMI). The use of the Robotics Lab equipment will give the student practical programming and troubleshooting skills used in the maintenance of automated systems.

Common Course Outcomes and Course General Education Outcomes:

Common Course Outcome	GE Outcome
The student shall explain the types and functions of generic PLC hardware.	Information Literacy Critical Thinking
The student will be able to describe the procedures for the troubleshooting PLC hardware.	Information Literacy Critical Thinking
The student will be able to demonstrate the use of the structure and organization of PLC program files for Studio 5000 and Siemens TIA.	Critical Thinking
The student will be able to demonstrate the use of the structure and organization of PLC program files for Studio 5000 and Siemens TIA.	Critical Thinking
The student will apply the basic relay instructions files for Studio 5000 and Siemens TIA in application programs.	Critical Thinking
The student will apply timers and counters instructions files for Studio 5000 and Siemens TIA in application programs.	Critical Thinking
The student will be able to demonstrate the construction of Human Machine Interface (HMI) objects with a PLC application programs.	Critical Thinking

The student will be able to analyze a problem with a systematic approach to troubleshoot an automated system.	Critical Thinking
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Course Goals:

Demonstrate the startup and use the editor functions for Studio 5000 and Siemens TIA.

Demonstrate and use of the structure and organization of PLC program files for Studio 5000 and Siemens TIA.

Demonstrate the use of the data structures for Studio 5000 and Siemens TIA.

Demonstrate and use the basic relay instructions files for Studio 5000 and Siemens TIA.

Demonstrate and use of timers and counters instructions files for Studio 5000 and Siemens TIA.

Demonstrate the construction of Human Machine Interface (HMI) with a PLC application program.

Tentative Course Schedule:

See Sign-Off Sheet for lab assignments.

Session	Class Activity	Tests
Class #1	Overview of Studio 5000 Software. Overview of SIEMENS TIA Software.	
Class #2	Studio 5000 Program/Memory Structures. SIEMENS Program/Memory Structures. Studio 5000 Relay Instructions. SIEMENS Relay Instructions.	
Class #3	Electrical Switches. Schematic Diagrams Symbols. Normally Closed.	
Class #4		
Class #5	Studio 5000 Force Functions. SIEMENS Force Functions.	
Class #6	Studio 5000 I/O Modules Configurations. SIEMENS I/O Modules Configurations.	
Class #7	Mid Term Review.	
Class #8		Mid Term
Class #9	Midterm Solutions. Studio 5000 Timers. SIEMENS Timers.	
Class #10	Studio Tasks. Studio 5000 Programs and Routines.	
Class #11	Studio 5000 Counters SIEMENS Counters. SIEMENS Functions. SIEMENS Function Blocks.	Click here to enter text.

Session	Class Activity	Tests
	SIEMENS Data Blocks.	
Class #12	SIEMENS Data Blocks Reusable.	
Class #13	Final Review	
Class #14	Lab	
Class #15		Final

FERPA Statement: Per the Family Educational Rights and Privacy Act (FERPA), college personnel are not allowed to release a student's personal information to anyone, including other students. For a full description of Oakland Community College's FERPA policy, please refer to: <https://www.oaklandcc.edu/student-services/college-policies/default.aspx>

Americans with Disabilities Act (ADA): Students requiring special assistance (including those affected by the Americans with Disabilities Act) should contact the ACCESS office, which will inform the instructor of any special accommodations pertaining to their learning.

Technology Appropriate Use Regulations: To access the Oakland Community College's policy on technology, please visit: <https://www.oaklandcc.edu/student-services/college-policies/default.aspx>

Public Safety: Oakland Community College has uniformed police officers at all academic locations. Please visit <https://www.oaklandcc.edu/student-services/college-policies/default.aspx>

College Closing and Cancelled Class Session:

Announcement of College Closings

To ensure the most effective and quickest way of finding out about a college or campus closing, be sure to enroll in OCC Emergency Alert at <http://www.oaklandcc.edu/ENS>. In the event of college or campus closings, an announcement will be made through major radio and television stations. *Note this does not include a cancelled session.*

Announcement of Cancelled Class Session

If a class session is to be cancelled, due to illness or other circumstances, an email will be sent to your college registered account. If you do not monitor your college e-mail account, you may want to forward or add to an e-mail program you frequently monitor.

Changes to Scheduled Content due to Closing or Cancellation.

Changes, if any, will be announced in the session following the closing or cancellation. DO NOT CALL OR E-MAIL REQUESTING INFORMATION ABOUT CHANGES.

Basis of Course Grade:

A	95 - 100	C	70 - 76
A-	92 - 94	C-	66 - 69
B+	89 - 91	D+	60 - 65
B	83 - 88	D	51 - 59
B-	80 - 82	F	50 -
C+	77 - 79		

WARNING: If you do not withdraw or make arrangements with the instructor to withdraw you from the course, *YOUR COURSE GRADE WILL BE DETERMINED FROM THE GRADES EARNED*. Student-initiated withdrawals will not be permitted beyond the twelfth week of instruction.

Grade Breakdown:

PLC Labs	35%
HMI Labs	15%
Mid-Term Exam	20%
Final Exam	30%

Incomplete: This faculty initiated mark will be used sparingly and only when an emergency prevents a student from completing course work during the regular College session. The student is responsible for completing a written agreement with the instructor detailing the requirements to be met for the completion of the "I" before it is assigned. This grade may be granted only when the student has completed at least 70% of the classwork with a passing grade of "C" or higher. The student is not to register for a course in which he or she has a current mark of "I". Without prior faculty-initiated action to change the "I", this mark will become a "WS" one year subsequent to its original issue. Marks of "I" do not satisfy prerequisites and are not transferable. This mark is not used in the calculation of GPA, but it may affect the eligibility for financial aid.

N Mark (Grade) Non-Attendant: This mark is awarded to students who, though registered, never attended class and did not officially drop. Oakland Community College is adopting the Federal policy language, which is:

- For synchronous classes (whether on-site or online), "attendance" is determined by attendance in a class, lecture, field activity, recitation, or laboratory activity.
- For online asynchronous classes, "attendance" is determined by submission of gradable work.

Marks of "N" do not satisfy prerequisites and are non-transferable. This mark is not used in the calculation of GPA, but it may affect eligibility for financial aid. Instructors confirm non-attendance immediately after the Census Date (found on the Section Deadline Dates website for each section, <https://www.oaklandcc.edu/section-dates>) and then report non-attendance to the

institution. Students who have received an "N" mark will not be permitted to enter the class, nor will the student be able to drop the class. After the "N" mark has been received, a change of grade will not be accepted for the student.

Attendance Policy: Attendance is mandatory for lecture and for the content delivered in the lecture. NO LECTURE WILL BE GIVEN FOR MISSED CONTENT.

Class Attendance / Tardiness: Students are expected to attend scheduled classes and to be on time. Failure to do this will be reflected in your final grade and in job recommendations. Attendance will be taken at the start of class, if the student is late, it is their responsibility to notify the instructor.

Online Course Site:

The online course site will be used for the following:

Accessing the PowerPoint slides used during lecture for the labs.

Labs Assignments:

Labs will be assigned on material that has been discussed in the lecture.

See LAB SIGN-OFF SHEET for assignments.

Labs are assigned points towards the course grade. See LAB SIGN-OFF SHEET for points assigned.

PLC LABS ARE TO BE CHECKED DURING CLASS. The program must be in run mode and the HMI running. Print the program to OneNote for your own record of the lab solution.

ALL LABS ARE DUE at the end of the fourteenth (14) session.

Midterm and Final:

THE MIDTERM AND FINAL EXAMS WILL BE BASED ON THE LABS.

NO MAKE UP WILL BE GIVEN FOR THE MIDTERM OR FINAL UNLESS YOU HAVE CONTACTED THE INSTRUCTOR FOR ALTERNATIVE ASSIGNMENTS.

Lab problems for both Midterm and Final are performance exams where programs will be assigned to be completed within the allotted time limit.

The Midterm and Final will begin at the specified start time for the time limit of the exam. Exam must be completed prior to the end of the scheduled class:

Midterm: **Session 8** in the T4 lab.

Final: **Session 15** in the T4 lab.

Testing policy: Exams can be made up if arrangements have been made by the student prior to the examination date.

Technical Issues:

If you cannot find the files location, the file was not loaded on the computer, or the file fails to open, notify the instructor immediately. If you fail to notify the instructor of any technical issue, it does not constitute a reason for not completing the lab assignment.

External Storage Devices:

External storage devices (USB drives) are permitted in the lab.

Back up your assignments at the end of each lab session.

It is highly recommended you create a duplicate of your work as a backup on your home computer.

External storage devices that are lost or damaged does not constitute an excuse for missing or uncomplete assignments.

Robotics Staff Contact:

Robotics Paraprofessional:

Richard Rymski

Room: T123

Phone: 248-232-4612

E-mail: rcrymski@oaklandcc.edu

Available Academic Assistance:

Academic Support Center (non-robotic assistance):

Hours 9AM-4PM, Room B-110

Phone 248-232-4435

Lab: Students must report any problems to OCC robotics staff.

Behavior: Disruptive or abusive behavior is not allowed in lectures or labs and subject to college discipline policy.

Students may be dismissed or failed in this class for violating safety regulations, failure to complete lab assignments, or failing their exams.

This syllabus is subject to revision as deemed necessary by the instructor. Appropriate notice shall be given to the students.